

Science-2: Mendel Quiz — Marking Scheme

06-April-2026 | Total: 20 Marks | Time: 20 min

Q1. Law of Segregation (2 marks)

2 marks — All three points present:

- Alleles **separate** from each other
- During **gamete formation** (meiosis)
- Each gamete receives **only one allele**

1 mark — Any two of the three points

0 marks — Vague, incorrect, or missing core idea

Q2. Law of Dominance (2 marks)

2 marks — All three points present:

- **Dominant** allele is expressed
- **Recessive** allele is masked/suppressed
- This occurs specifically in a **heterozygote**

1 mark — Missing the heterozygote context

0 marks — Incorrect

Q3. Law of Independent Assortment (2 marks)

2 marks — All three points present:

- **Different genes** (or pairs of alleles)
- Assort **independently** of one another
- During **gamete formation**

1 mark — Any two components present

0 marks — Incorrect

Q4. Test Cross (6 marks)

(a) Definition — max 2 sentences (2 marks)

2 marks — States: cross between an organism with the **dominant phenotype** (of **unknown genotype**) and a **homozygous recessive** organism

1 mark — Mentions dominant × recessive but omits 'homozygous recessive' or 'unknown genotype'

0 marks — Incorrect

(b) Use — 1 sentence (1 mark)

1 mark — To **determine the genotype** (AA vs Aa) of the dominant-phenotype parent

0 marks — Incorrect or too vague

(c) Punnett Square — Cross: Aa × aa (3 marks)

Test Cross: Aa × aa

	a	a
A	Aa	Aa
a	aa	aa

Result: 1 Aa : 1 aa → 1 dominant phenotype : 1 recessive phenotype

3 marks — Correct cross (Aa × aa), gametes labelled on headers, all four offspring correct (Aa, Aa, aa, aa)

2 marks — Correct offspring but gametes missing from headers, OR one minor filling error

1 mark — Correct cross attempted but major filling errors

0 marks — Missing, or wrong cross (e.g. AA × aa)

■ If AA × aa is used: all offspring are Aa, which does not demonstrate a test cross — lose Punnett marks; keep definition/use marks.

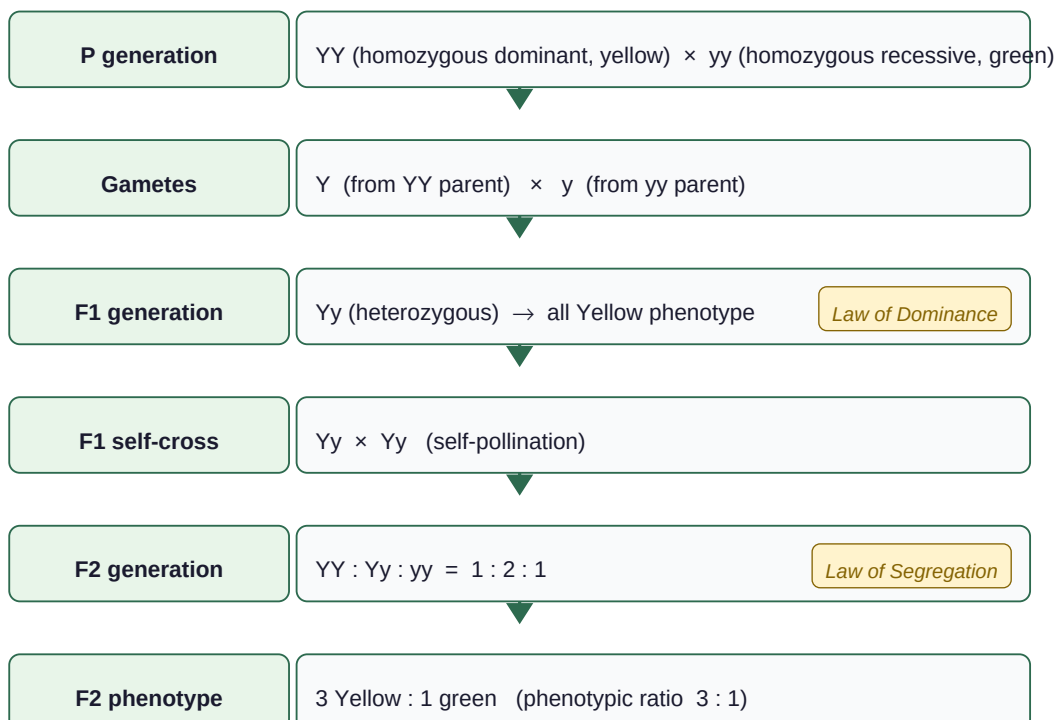
Q5. Classic Mendel Experiment — Seed Colour (8 marks)

■ Examiner Note: The question states 'seed shape' but describes colour alleles (Yellow/green). Award full marks based on seed colour. This appears to be a typo; do not penalise students who correctly address seed colour.

(a) Schematic / Flow Diagram with Technical Jargon

Students must draw a flow diagram showing all steps below, using the correct technical terms (homozygous dominant/recessive, heterozygous, gametes, F1, F2, phenotype, genotype, genotypic/phenotypic ratio, self-pollination). Laws must be labelled at the correct step.

Expected schematic (model answer):



(b) P Generation — 1 mark

1 mark — Correctly states **YY × yy**

0 marks — Incorrect parental genotypes

(c) Gametes — 1 mark

1 mark — States gametes are **Y** (from YY) and **y** (from yy)

0 marks — Incorrect

(d) F1 Generation — 2 marks

2 marks — States both **genotype (Yy)** AND **phenotype (all yellow)**

1 mark — Genotype only OR phenotype only

0 marks — Incorrect

(e) F1 Self-Cross — 1 mark

1 mark — Correctly writes **Yy × Yy**

0 marks — Incorrect

(f) F2 Punnett Square — 2 marks

Cross: **Yy × Yy** — standard 2×2 grid, gametes labelled on column/row headers:

F2 Punnett Square: Yy × Yy

	Y	y
Y	YY	Yy
y	Yy	yy

Genotypic ratio: 1 YY : 2 Yy : 1 yy | Phenotypic ratio: 3 Yellow : 1 green

2 marks — Correct 2×2 grid, gametes (Y, y) labelled on top row AND left column, all four offspring correct (YY, Yy, Yy, yy)

1 mark — Offspring mostly correct but gametes missing from headers, OR one filling error

0 marks — Missing, wrong cross, or majority incorrect

■ Punnett Square format: Gametes **MUST** appear as column/row headers — not just offspring cells. Missing headers → deduct 0.5 mark.

(g) Ratios — 1 mark

1 mark — States **BOTH** ratios:

- Genotypic: **1 YY : 2 Yy : 1 yy**

- Phenotypic: **3 yellow : 1 green**

0.5 marks — Only one ratio given

0 marks — Both incorrect

(h) Laws Identified — 1 mark

Step	Law
F1 generation (all yellow despite Yy)	Law of Dominance
F2 generation (alleles separate → 3:1 ratio)	Law of Segregation
Independent assortment	NOT applicable in a monohybrid cross (credit if student correctly states N/A)

1 mark — Both F1 and F2 laws correctly matched

0.5 marks — Only one law correctly placed

0 marks — Both incorrect, or laws placed at wrong steps

General Evaluation Rules

- Evaluate **only within sentence limits**; ignore content beyond the given limit.
 - If the **first sentence is incorrect**, do not award credit for later corrections in the same answer.
 - **Missing Punnett square** where required → **0 marks for that part**, regardless of written answer.
 - **Minor notation errors** (e.g., lowercase for dominant allele) → deduct **0.5 marks**.
 - **Wrong cross in Q4** (AA × aa instead of Aa × aa) → lose Punnett marks; keep definition/use marks.
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Mark Summary

Q	Topic	Marks
1	Law of Segregation	2
2	Law of Dominance	2
3	Law of Independent Assortment	2
4	Test Cross	6
5	Classic Mendel Experiment	8
Total		20